

```
1 import java.util.*;
2
3 public class Main {
4
5     static class Item {
6         int value;
7         int weight;
8
9         Item(int value, int weight) {
10             this.value = value;
11             this.weight = weight;
12         }
13     }
14
15     public static void main(String[] args) {
16         Scanner sc = new Scanner(System.in);
17
18         // Number of items
19         System.out.println("Enter the number of items:");
20         int n = sc.nextInt();
21
22         Item[] items = new Item[n];
23
24         // Input value and weight
25         for (int i = 0; i < n; i++) {
26             int value = sc.nextInt();
27             int weight = sc.nextInt();
28             items[i] = new Item(value, weight);
29         }
30     }
```

```

31 // Knapsack capacity
32 int capacity = sc.nextInt();
33
34 // Sort according to value/weight ratio
35 Arrays.sort(items, (a, b) -> {
36     double r1 = (double) a.value / a.weight;
37     double r2 = (double) b.value / b.weight;
38     return Double.compare(r2, r1);
39 });
40
41 double maxProfit = 0;
42
43 // Select items greedily
44 for (int i = 0; i < n; i++) {
45
46     if (capacity >= items[i].weight) {
47         maxProfit += items[i].value;
48         capacity -= items[i].weight;
49     }
50     else {
51         double ratio = (double) items[i].value / items[i].weight;
52         maxProfit += ratio * capacity;
53         capacity = 0;
54         break;
55     }
56 }
57
58 System.out.println("Maximum Profit = " + maxProfit);
59
60 sc.close();
61 }

```

Enter the number of items:

3

60 10

100 20

120 30

50

Maximum Profit = 240.0

...Program finished with exit code 0

Press ENTER to exit console.